

One model or two? Live on-device speech recognition in English and Spanish on a mid-range Android phone

Daniel Valcarce

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Code, corpus builder and every result: github.com/davamix/ondevice-streaming-asr-bench, tag v1.0

ABSTRACT

An app that transcribes speech live, entirely on the phone, in English and Spanish, can ship one model for both languages or one model per language. English has strong open streaming models; Spanish had no deployable streaming model when this study began. We price that asymmetry on a 2021 mid-range phone (Snapdragon 870, 6 GB of RAM). Five deployable stacks were compared: Android’s on-device recognizer, Moonshine streaming (English tiny, small, medium; Spanish small), Moonshine base-es, Parakeet TDT 0.6b v3 and Whisper base and small. All heard identical audio paced to the wall clock: 100 clips per source from FLEURS English and Spanish and LibriSpeech test-other, plus continuous six-minute sessions and a live-microphone check. One model suffices. Parakeet reaches 4.47% word error rate (WER) in Spanish, about 40% fewer errors than any per-language option (7.2–7.4%), and is level with the best English model on clean speech (4.97% against 4.71%). It sustains 0.6 of real time over six minutes of continuous speech without drift. It costs 670 MB of disk and about 1.1 GB of memory, and without half a second of silence padding per segment it drops whole sentences. Moonshine small in English and Spanish (346 MB) is the smaller, cooler alternative, with Spanish no better than the free platform recognizer’s. That recognizer, in turn, misses one word in four on noisy English. A live microphone reproduced the file-fed method’s text exactly, and showed its latencies to be about 0.2 s optimistic.

Keywords: on-device speech recognition; streaming ASR; live transcription latency; Android; Spanish; Parakeet; Moonshine; Whisper; benchmarking methodology

1 Introduction

A speech-to-text feature that runs entirely on the phone avoids the network, the server bill, and the question of sending a user’s voice elsewhere. Making it *live*, with text appearing while the user speaks, changes what a good model is. A model with a real-time factor of 0.3 can still feel slow if it needs the whole utterance before it emits anything. It feels broken if it keeps rewriting text the user has already read. The numbers that decide a live product are latency-shaped: how soon text appears after speech starts, how soon it is final after speech stops, how often displayed text is revised, and whether the model keeps up with the microphone over minutes rather than seconds.

This study asks one product question, for one app that must transcribe English and Spanish:

Can one model serve both English and Spanish at acceptable live latency on a mid-range phone, or do we ship a different model per language?

The two languages are not in the same position. English has strong open streaming models: Moonshine’s streaming models [5] cover English from 78 MB. When the study began, no deployable streaming model for Spanish was known. sherpa-onnx [8] offers no Spanish streaming transducer, and Moonshine’s Spanish streaming weights appeared to exist only in a training format. They turned out to be on the vendor’s content network (§2.2). The Spanish options were the platform’s own recognizer, or offline models made live by cutting the audio with a voice activity detector (VAD): Parakeet TDT 0.6b v3 [4, 6], which covers 25 European languages, and Whisper [3]. The experiment exists to price that asymmetry. The size budget was deliberately left open, so that the accuracy-against-megabytes curve could inform it.

The contributions are:

- a controlled comparison of five deployable stacks (a model plus the runtime that executes it), nine configurations in all, on one Snapdragon 870 phone, in English and Spanish, on clean and noisy speech, with byte-identical audio paced to the wall clock (§3, §4);
- an answer: one model suffices, and it is Parakeet, with its costs and those of the per-language alternative quantified (§6);

- continuous six-minute sessions, which show that no surviving stack falls behind real time or throttles, and a live-microphone check of the file-fed method, which measures its timing bias at about 0.2 s (§4.4, §4.5);
- eleven measurement and integration pitfalls met during the study, most of which changed a number that had already been published (§5);
- the harness, corpus builder, scorer and every result file, pinned to model and dataset revisions (Appendix B).

2 Setup

2.1 Device

Table 1. Device under test.

Device	Xiaomi M2012K11AG (Poco F3 / Mi 11i)
System on chip	Qualcomm SM8250, Snapdragon 870
CPU	1× Cortex-A77 at 3.19 GHz, 3× A77 at 2.42 GHz, 4× A55 at 1.80 GHz
Memory	6 GB, of which about 2 GB typically available
Software	Android 13 (API 33), arm64-v8a

API 33 matters twice. It provides `SpeechRecognizer.createOnDeviceSpeechRecognizer`, a recognizer the app does not have to ship, and `EXTRA_AUDIO_SOURCE`, which lets that recognizer be fed from a file instead of the microphone.

The phone is its owner’s personal, daily-use handset, and every run followed a safety policy. Runs start only below 35 °C battery temperature, unplugged, with the battery between 30 and 80%, and abort at 43 °C. Continuous inference is capped at 10 minutes, with 2-minute breaks. Nothing is rooted, and thermal throttling is never disabled. The last rule is also a methodological choice: live transcription runs hot for minutes, so throttling is part of what is being measured. An emulator was used to test the harness only; no number reported here comes from it.

2.2 Stacks under test

The unit of comparison is the deployable stack, because the runtime cannot be held constant across these options. Moonshine ships its own SDK, the platform recognizer is a black box running in another process, and sherpa-onnx runs the rest. Differences are therefore attributed to stacks, not to models.

Table 2. The five stacks. “VAD + re-decode”: an offline model made live as described in §3.2.

Arm	Stack	How it is live	Languages	Disk	Licence
A	Android on-device recognizer (API 33)	native streaming	en, es (language packs)	0 MB	platform
B	Moonshine v2 streaming tiny, small, medium (en); small (es); SDK moonshine-voice 0.1.5	native streaming	one per model	78 / 224 / 416 MB; es 122 MB	MIT
C	Moonshine base-es, Moonshine SDK, Silero VAD	VAD + re-decode	es	65 MB	non-commercial
D	Parakeet TDT 0.6b v3, int8, sherpa-onnx 1.13.8, Silero VAD	VAD + re-decode	25, incl. en, es	670 MB	CC-BY-4.0
E	Whisper base and small, multilingual, int8, sherpa-onnx 1.13.8, Silero VAD	VAD + re-decode	en, es (told which)	161 / 375 MB	MIT

Arm E is a calibration baseline: Whisper expects 30-second windows and is not built for live use. Two options were excluded before testing. Vosk has no punctuation or casing and is weak on noisy speech. Whisper large-v3-turbo carries about 1.6 GB of weights against about 2 GB of available memory. Moonshine’s Spanish streaming model was first believed not to exist in a deployable format. It was found on the vendor’s content network, where the SDK’s own model catalogue fetches it, and was added in the fourth of five phases. Every model is loaded from disk at a pinned revision, and nothing downloads itself (Appendix B).

3 Method

3.1 Paced file-fed streaming

Every measured run is fed from a file, never from a microphone: nobody can say the same sentence twice identically, so microphone input would confound every comparison with the speaker’s delivery. To make a file behave like a microphone, 16 kHz mono 16-bit audio is released in 100 ms frames on the wall clock, and frame k is not handed over before $k \cdot 100$ ms have passed. Every stack hears byte-identical input, repeatably. Arm A receives the frames through a pipe via `EXTRA_AUDIO_SOURCE`, with a fresh recognizer per clip; Arm B receives them through the Moonshine SDK’s `addAudio()`; Arms C–E feed them to a VAD.

The feeder releases frame k at the start of its 100 ms slot, while a microphone can deliver it only at the end, so every stack receives audio up to one frame early. The head start is the same for every stack and does not affect comparisons. §4.5 measures its effect on absolute latency.

3.2 Offline models made live

Arms C, D and E follow the pattern of sherpa-onnx’s own simulated-streaming example. Silero VAD [9] (threshold 0.5, minimum silence 0.25 s, minimum speech 0.25 s) cuts the paced audio into utterances, and each closed segment is decoded once for its final text. While speech is still open, the audio so far is re-decoded every 500 ms and shown as partial text. That is the Moonshine SDK’s update interval, so both kinds of stack refresh text on the same cadence. A partial is skipped if the previous decode is still running, and decoding runs on its own thread, so the feed never stalls. The VAD’s maximum speech duration is 20 s; the Kotlin API’s default of 5 s would split most of the corpus’s sentences at a breath. The sherpa-onnx arms use four inference threads. The Moonshine SDK exposes no thread setting, so ONNX Runtime’s default applies to Arms B and C.

3.3 Corpus

Table 3. Corpus. Short clips are one utterance each; the sessions are continuous speech.

Source	Language	Condition	Clips	Reference words	Length
FLEURS en_us, test [1]	English	clean read speech, level-matched	100	1,710	3–8 s
LibriSpeech test-other [2]	English	noisy read speech	100	1,489	3–8 s
FLEURS es_419, test [1]	Spanish	clean read speech	100	1,767	3–10 s
Session, English	English	34 FLEURS utterances, level-matched	1	772	6.1 min
Session, Spanish	Spanish	27 FLEURS utterances	1	719	6.2 min

FLEURS was chosen so that English and Spanish would share a corpus, and LibriSpeech test-other adds the noisy case FLEURS lacks. FLEURS’s English recordings turned out to be about 40 dB quieter than its Spanish ones (§5.1). Every English FLEURS figure here uses a level-matched copy, each clip lifted by one static gain to -23 dBFS RMS, unless it is marked “very quiet”. Sessions are built by joining unused clips with 0.5–1.5 s of silence, which gives continuous speech with an exact reference transcript. Clip selection is seeded, and every rebuild reproduced every earlier file byte for byte.

3.4 Metrics

- **WER and CER**, scored on a PC with jiwer [11]; the phone emits text only. Normalisation lowercases, strips punctuation and spells out digits, reading English four-digit numbers from 1100 to 1999 as years, and clock times, percentages and thousands as spoken. It keeps Spanish accents. Corpus WER is total edits over total reference words. An empty hypothesis counts every reference word as deleted, and each clip counts once however many runs measured it.
- **Final text**: time from the end of the audio to the final text, clamped at zero, so text already final when the audio ends counts as 0. In sessions it is taken per utterance, from the end of that utterance.
- **First text**: time from speech onset to the first text shown. Onset is estimated on the PC as the first 20 ms frame within 25 dB of the clip’s loudest.
- **Revisions**: words already shown that were later changed or removed, from the longest common prefix of successive hypotheses. Pure extension costs nothing.
- **Compute share**: time the stack spent computing, on the feed path and on decode threads, partials included, over the audio’s duration. Above 1.0 a stack cannot keep up with a microphone. Over a continuous session this is the sustained real-time factor, `rtf_sustained`; sessions also record it per 30 s of audio.

- **Peak memory:** the process’s resident high-water mark, harness included. Arm A’s recognizer runs in Google’s process, so only the harness’s own ~115 MB is visible for it.
- **Battery points and battery temperature.** Battery temperature is the only temperature this phone exposes to an app, and it can lag by minutes (§5.11).

3.5 Protocol and statistics

Runs used two to four repetitions. The first was discarded as a cold start, and arm order was randomised per repetition. The VAD arms and Moonshine’s Spanish model gave identical text in every repetition, so repetitions add evidence about timing, not accuracy. Accuracy rests on the number of clips, which is why each source grew from 20 to 100 (§5.5). WER differences between two stacks are tested with a paired bootstrap over clips and reported as 95% intervals. Latencies are medians. Sessions ran once per language per stack, so they are indicative. A drift over a session is tested with a Spearman rank correlation against position and a permutation test.

4 Results

4.1 Accuracy

Table 4. Word error rate, with character error rate in parentheses, per stack and condition. 100 clips per source unless marked. Best per column in bold.

Stack	Disk	Spanish	English, clean	English, noisy
A Platform recognizer	0 MB	7.39% (3.55)	9.72% (5.22)	27.09% (20.11)
B Moonshine tiny-en	78 MB	—	11.25% (5.76)*	14.43% (6.44)*
B Moonshine small-en	224 MB	—	7.36% (3.01)	8.92% (4.19)
B Moonshine medium-en	416 MB	—	4.71% (1.73)	7.02% (3.58)
B Moonshine small-es	122 MB	7.24% (2.27)	—	—
C Moonshine base-es	65 MB	5.09% (2.05)	—	—
D Parakeet TDT 0.6b v3	670 MB	4.47% (1.76)	4.97% (2.30)	9.34% (5.57)
E Whisper base	161 MB	14.73% (4.35)*	11.88% (6.64)*	22.82% (9.62)*
E Whisper small	375 MB	11.24% (4.22)*	5.31% (2.39)*	14.43% (6.19)*

* The original 20 clips per source only. On the very quiet original FLEURS English recordings (20 clips) the platform recognizer scores 9.69%, Moonshine small 27.29%, medium 25.31%, Whisper small 10.31% and Parakeet 14.69% (§5.1).

Spanish. Parakeet is the most accurate option, at 4.47%. It beats the platform recognizer by 2.9 points and Moonshine’s Spanish streaming model by 2.8, both resolved (Table 5): about 40% fewer errors than any per-language option. Moonshine small-es is level with the platform recognizer (7.24% against 7.39%). Moonshine publishes 4.9% for it, on different data, decoding whole utterances; that did not carry over to this corpus and pipeline. Arm C is level with Parakeet at a tenth of the disk, but it is not live (§4.2), and its licence forbids shipping it.

English. Moonshine medium is the most accurate English model (4.71% clean, 7.02% noisy). Parakeet is level with it on clean speech and 2.3 points behind on noisy speech, a gap the data do not resolve. Much of that gap is one failure mode, which padding fixes (§5.7). The platform recognizer is competitive on clean speech (9.72%), but it misses about one word in four on noisy speech (27.09%), 18 points worse than Moonshine small.

Table 5. Paired comparisons of WER (percentage points; positive means the first stack makes more errors), paired bootstrap over clips, 100 clips.

Comparison	Difference	95% interval	Verdict
Spanish			
Platform – Parakeet	+2.9	+0.6 to +5.8	Parakeet better
Moonshine small-es – Parakeet	+2.8	+1.3 to +4.3	Parakeet better
Moonshine small-es – base-es (C)	+2.2	+0.5 to +3.8	C better
Platform – Moonshine small-es	+0.1	–2.7 to +3.5	not resolved
Platform – base-es (C)	+2.3	–0.0 to +5.1	not resolved, only just
base-es (C) – Parakeet	+0.6	–0.5 to +1.8	not resolved
English			
Platform – Moonshine small, noisy	+18.2	+12.9 to +23.7	Moonshine better
Platform – Moonshine small, clean	+2.4	+0.1 to +5.1	Moonshine better, barely
Platform – Parakeet, clean	+4.8	+2.6 to +7.4	Parakeet better
Moonshine small – medium, clean	+2.7	+1.5 to +3.9	medium better
Moonshine small – medium, noisy	+1.9	+0.0 to +3.8	medium better, barely
Moonshine small – Parakeet, clean	+2.4	+1.3 to +3.6	Parakeet better
Moonshine small – Parakeet, noisy	–0.4	–2.8 to +1.7	not resolved
Moonshine medium – Parakeet, clean	–0.3	–1.2 to +0.7	not resolved
Moonshine medium – Parakeet, noisy	–2.3	–5.1 to +0.2	not resolved

4.2 Latency

Table 6. Latency medians. Final text: from the end of the audio, clamped at zero. First text: from speech onset. Revisions: median words rewritten per clip. Values are Spanish / clean English / noisy English where a stack covers them.

Stack	Final text, ms	First text, ms	Revisions
A Platform recognizer	0 / 69 / 81	711 / 730 / 752	6 / 8 / 8
B Moonshine tiny-en	— / 0 / 84	— / 745 / 625	— / 16 / 10
B Moonshine small-en	— / 222 / 486	— / 968 / 818	— / 8 / 8
B Moonshine medium-en	— / 539 / 772	— / 996 / 810	— / 7 / 8
B Moonshine small-es	0 / — / —	738 / — / —	14 / — / —
C Moonshine base-es	2504 / — / —	1328 / — / —	8 / — / —
D Parakeet	0 / 250 / 349	980 / 1528 / 1366	12 / 12 / 8
E Whisper base	471 / 618 / 678	1098 / 1196 / 1133	14 / 9 / 16
E Whisper small	2750 / 2683 / 2424	1721 / 1833 / 1787	2 / 4 / 5

From a live microphone, add about 0.2 s to every final and first text figure (§4.5). Medians are from the runs the project README reports; the pooled medians over every run (results/README.md) differ by at most a few tens of milliseconds.

Final text is quick for every live stack. It takes 0–350 ms for Parakeet and 0–81 ms for the platform recognizer. Moonshine takes 0–772 ms, rising with model size, because each model pass takes about 200, 450 and 550 ms for tiny, small and medium. Arm C and Whisper small take 2.4–2.8 s: their decodes are slower than the 500 ms partial interval, so the decode thread never idles while someone speaks, and final decodes queue behind partials.

First text is where the VAD arms are weakest. From speech onset, Parakeet shows text in 1.0–1.5 s, against 0.6–1.0 s for Moonshine and about 0.7 s for the platform recognizer. A partial has to wait for the VAD to declare speech (at least 0.25 s of it), then one 500 ms interval, then a decode. Moonshine’s first text is paced by its SDK’s 0.5 s update interval, not by its roughly 80 ms of algorithmic lookahead: 108 of 114 first-text times for Moonshine tiny fell within 100 ms of $40 + k \cdot 500$ ms. Spanish is not slower to show text on any stack; it only looked so when timed from the start of the clip (§5.6).

4.3 Compute, memory and heat

Table 7. Cost per clip. Compute is the share of real time spent computing with 500 ms partials, and with final text only in parentheses. Battery is per 240 clips; cooling pauses are counted per run, with the run’s length.

Stack	Compute	Peak memory	Battery	Cooling pauses (run)
A Platform recognizer	—	— (in Google’s process)	2–4 points	0 (39–64 min)
B Moonshine tiny-en	0.40–0.42	363–475 MB	6 points	0 (39 min)
B Moonshine small-en	0.72–0.73	605–760 MB	10 points	0–2 (39–111 min)
B Moonshine medium-en	0.80–0.82	933–967 MB	11–13 points	1–7 (44–132 min)
B Moonshine small-es	0.43	640 MB	9 points	0 (61 min)
C Moonshine base-es	1.16 (0.45)	872 MB	28 points	8 (78 min)
D Parakeet	0.52–0.62 (0.20–0.23)	1027–1034 MB	10–13 points	0–3 (41–81 min)
E Whisper base	0.76–0.88 (0.27–0.30)	572 MB	15 points	7 (87 min)
E Whisper small	1.24–1.36 (0.52–0.59)	997 MB	29 points	15 (156 min)

Peak memory is a process high-water mark that includes the harness. A cooling pause is a clip that had to wait for the 35 °C start gate. Ranges span the runs made with each stack.

Showing live text costs more than the model. Final decodes alone need 0.20–0.23 of real time for Parakeet, and re-decoding the open utterance every 500 ms raises that to 0.52–0.62. Whisper small goes past 1.0 (§5.8). **Memory did not bite.** Parakeet had been flagged in advance as the stack most likely to fail on memory, with 1.5–2 GB expected. It peaked at about 1.03 GB per clip, loaded in 2.6 s, and neither crashed nor was killed in more than 900 clips. **Heat follows compute.** Whisper small paused to cool 15 times in 156 minutes and Arm C 8 times in 78, while Moonshine small-es never passed 31.7 °C.

4.4 Sustained six-minute sessions

A stack can keep up clip by clip and still fall behind the microphone once the phone is hot, and then never catch up. The stacks that survived the per-clip comparison were therefore run on one continuous session per language: 34 English utterances in 6.1 minutes and 27 Spanish ones in 6.2, fed in real time, one stack per run, with a 2-minute break between the two sessions. The phone was unplugged and idle on its home screen, and its system log confirms nobody used it during these runs (§5.10). Every 30 s of audio the harness recorded compute, feeder delay, battery temperature, memory and phone use, and for every final segment it recorded when the text was committed.

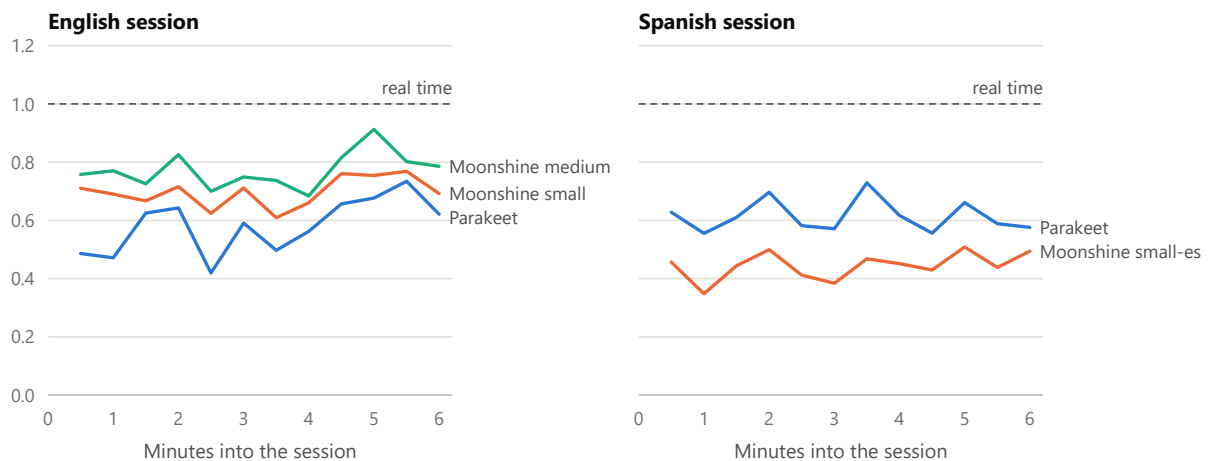


Figure 1. Share of real time spent computing, per 30 s of audio, across one continuous session per language. Above 1.0 a stack falls behind the microphone for good. None came near it.

Table 8. Six-minute sessions. “Drift” asks whether final-text lag rises or falls with position in the session (Spearman rank correlation, permutation test).

Session	Compute, session (per clip)	Worst 30 s	Final text, median / p90	Drift	Temperature	WER
Parakeet, English	0.59 (0.62)	0.73	0 / 366 ms	none (p = 0.27)	32.5 → 35.0 °C	11.14%*
Parakeet, Spanish	0.61 (0.56)	0.73	23 / 416 ms	none (p = 0.24)	34.2 → 35.7 °C	6.26%
Moonshine small-en	0.70 (0.73)	0.77	0 / 646 ms	none (p = 0.97)	33.0 → 35.2 °C	7.25%
Moonshine small-es	0.45 (0.43)	0.51	0 / 178 ms	none (p = 0.47)	33.2 → 33.5 °C	8.62%
Moonshine medium-en	0.78 (0.81)	0.91	220 / 919 ms	none (p = 0.41)	33.2 → 36.7 °C	6.74%

* 8.3% had one sentence that came back empty been transcribed (§5.7). The platform recognizer scored 6.54% on the same Spanish session. Session sentences are not the short clips’ sentences, so session WER compares stacks on the same audio, not with Table 4.

Nothing falls behind. Over six continuous minutes every stack used what it used per clip, within 0.05 of real time, and its worst 30 seconds stayed well below real time. Moonshine stalls its feeder by up to 1.6 s while it decodes, because it decodes inside `addAudio()`, but the delay did not accumulate. **Nothing throttled.** Final-text lag shows no trend in any session. Parakeet’s English compute does rise through its session, from about 0.5 to 0.7, but that is the audio: its later sentences are longer, and every partial re-decodes everything said so far. Its decode speed per second of audio actually improved slightly (105 to 92 ms), and its Spanish session, run next on a warmer phone, is flat. The platform never reported a thermal status above “none”. **Memory creeps up.** Every stack ended a session 35–170 MB above its level at 30 seconds, and Parakeet reached 1.11 GB across its two sessions. The growth is not steady (Moonshine medium’s fell back by 100 MB mid-session), and its cause was not established.

4.5 The real-microphone check

Every measured run rests on the claim that pacing a file shows a model what a live microphone would. The check tests it directly. The phone’s owner read six corpus sentences aloud while one stack transcribed the microphone live (`AudioRecord`, `VOICE_RECOGNITION` source, the same 100 ms frames). The recording was kept, and the same samples then went through the same stack again, file-fed exactly as in every measured run. There were four takes of 60 s: Parakeet and Moonshine small, in English and in Spanish.

Table 9. Live microphone against file-fed, same recording, same stack. Compute and feeder delay are given as live / file-fed.

	Parakeet, en	Parakeet, es	Moonshine small-en	Moonshine small-es
Text	identical	identical	2 of 82 words differ	identical
Compute	0.39 / 0.37	0.34 / 0.33	0.54 / 0.54	0.31 / 0.31
Feeder delay, worst	20 / 20 ms	19 / 28 ms	923 / 798 ms	517 / 498 ms
Segments	identical	identical	differ slightly	differ slightly
Final text later live (median)	+210 ms	+208 ms	+132 ms	+256 ms

The simulation holds for text and compute. Parakeet produced byte-identical text from the live microphone and from the file, and so did Moonshine in Spanish. Moonshine’s English differed by two words, within the small repetition-to-repetition variation its English streaming already showed. Compute matched within 0.02 everywhere. **Live text arrives about 0.2 s later.** With identical segments, Parakeet’s VAD closed each one 0.19–0.20 s later relative to the audio, and its text committed 0.21 s later. About 0.1 s of that is the feeder’s one-frame head start (§3.1), and the rest is the phone’s capture path. Every first-text and final-text figure in this paper is therefore about 0.2 s optimistic for a live microphone, equally for every file-fed stack. **This microphone is quiet.** Speech reached the model at –39 to –42 dBFS RMS, about 17 dB below the level-matched corpus and 20 dB above the very quiet FLEURS English, and Moonshine produced text for every line. The takes are the owner’s voice; they stay private and are not scored.

4.6 Trade-offs

Figures 2–4 draw the per-clip results of §4.1–4.3 as costs against accuracy, one panel per condition. Lower and further left is better. Blue is Parakeet, orange Moonshine’s streaming models, gray the other stacks; hollow points were measured on 20 clips per source, filled ones on 100. The recommendation in §6 reads off them: in Spanish, only Parakeet sits low; in English, Moonshine medium and Parakeet share the bottom, at different sizes; and nothing bundled is as cheap as the platform recognizer, which fails on noisy English.

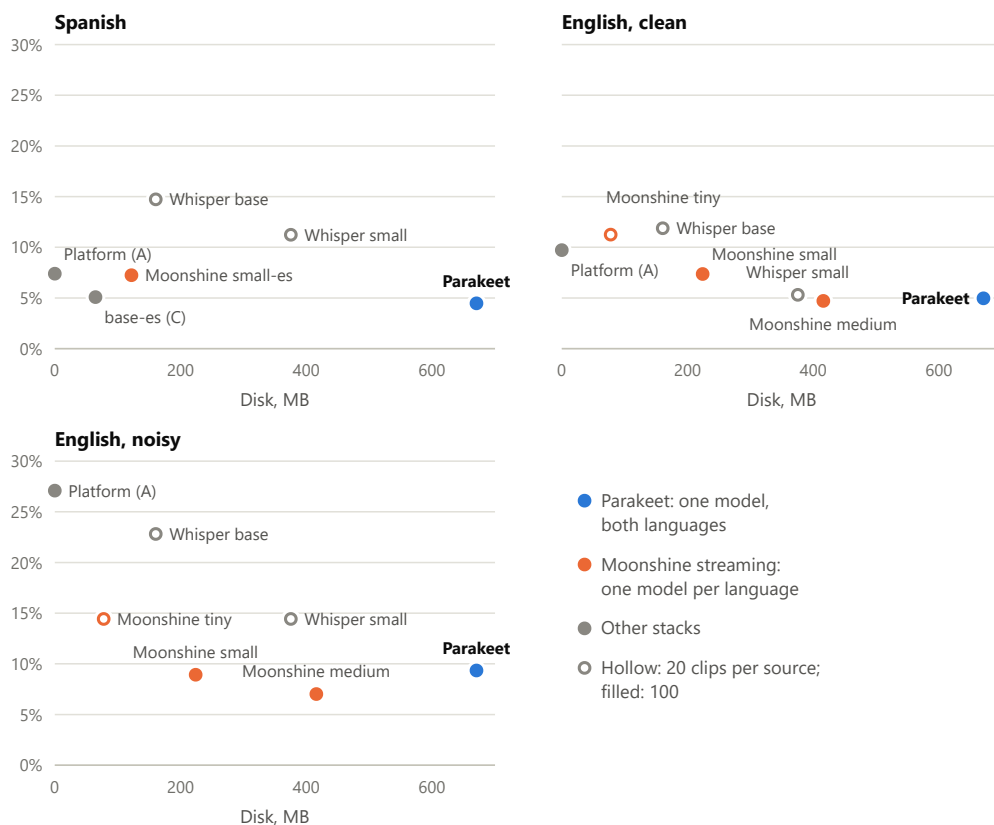


Figure 2. Accuracy against size on disk. Lower and further left is better. Parakeet (670 MB) is the most accurate in Spanish and near the most accurate in English. The platform recognizer (0 MB) is competitive except on noisy English.

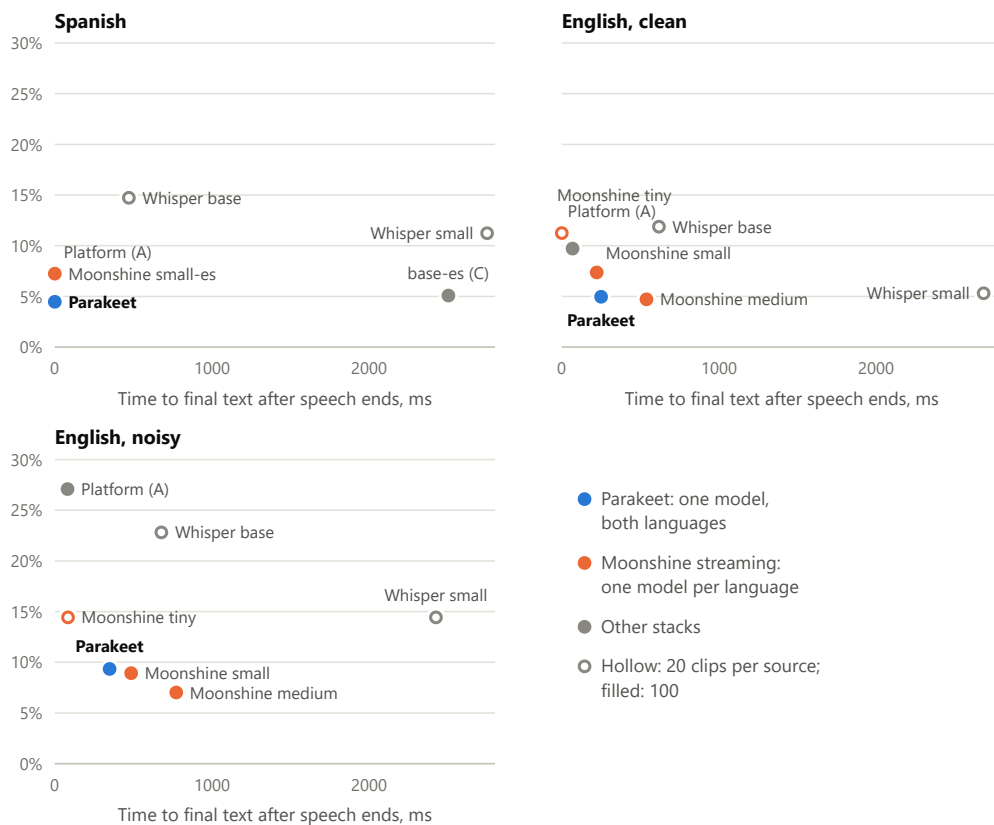


Figure 3. Accuracy against time to final text after speech ends. Parakeet, Moonshine and the platform recognizer finish within 0.8 s. Arm C (base-es) and Whisper small take 2.4–2.8 s.



Figure 4. Accuracy against peak memory, per model. The platform recognizer is not shown: it runs in Google’s process. The horizontal axis starts at 300 MB.

5 Pitfalls

Most of the following changed a number that had already been published. None of the eleven is exotic, which is the reason for listing them.

5.1 Recording level is a variable

FLEURS was chosen so that English and Spanish would be “the same corpus, recorded the same way”. On loudness they are not: the English test recordings sit at a median of -62.6 dBFS RMS, against -22.2 for Spanish and -23.6 for LibriSpeech. The difference is inherited from the dataset. The platform recognizer copes (9.69% on the quiet clips). Moonshine returns no text, and no error, on some of them: Moonshine tiny left 15 of 120 rows blank and scored 30.05%. The same clips lifted by a static gain left none blank and scored 11.25%. The six-minute English session, built before this was known, had 25 of its 34 utterances below -50 dBFS, and was level-matched before any session was run.

5.2 An empty transcript is every word missed

The first scorer skipped rows that returned no text. That removed their words from the denominator, so a stack that heard nothing scored better for it: Moonshine tiny’s pilot scored 19.3% that way and 26.9% in fact.

5.3 Filtering “bad” rows can flatter what is measured

It is tempting to drop rows in which the feeder fell behind schedule. But that delay is the recognizer refusing input, and it happens most on the audio the recognizer finds hardest. In one run, dropping those rows made the platform recognizer’s final-text latency on noisy English look 2.8 times better (315 to 114 ms), by keeping only the easy third. Headline figures use every row.

5.4 Formatting is not recognition

The scorer read “1848” as “one thousand eight hundred forty eight”, while LibriSpeech’s references say “eighteen forty eight”. Stacks that write years as digits lost about four WER points on noisy English (the platform recognizer: 33.33% to 29.31% on 20 clips once fixed). Clock times, percent signs and thousands separators followed. Each rule applies to reference and hypothesis alike.

5.5 Twenty clips could not rank the stacks

Deterministic stacks produce the same text in every repetition, so four repetitions of 20 clips are 20 clips of evidence. On 20 Spanish clips, Parakeet's lead over the platform recognizer was +1.55 points with a 95% interval of -2.70 to +5.18; on 100 clips it is +2.91, from +0.55 to +5.81. Some 20-clip point estimates between English models reversed on 100 clips.

5.6 Measure from the event the metric is named after

Every stack looked 0.6–0.9 s slower to show Spanish text than English. The FLEURS Spanish clips open with a median 1.31 s of silence, against 0.32–0.46 s for the English sources, and first text was timed from the start of the clip. From speech onset, the platform recognizer shows text in about 0.7 s in both languages.

5.7 Parakeet can return nothing for a VAD segment

On some segments Parakeet's final decode is empty, although partial text was on screen moments earlier. Silero VAD starts a segment close to speech onset, and Parakeet then receives audio that begins almost mid-syllable. A PC reproduction of the same pipeline, with the same model files, shows what padding does: half a second of leading silence moves noisy English from 9.47% to 7.12%, clean English from 5.26% to 4.68% and Spanish from 4.13% to 4.02%. Empty segments fall from 8 to 2. Trailing silence alone makes things worse (12.09%). In the English session, one 6.9 s segment of 55 came back empty: a whole 22-word sentence. With the padding it decodes word for word. Padded Parakeet was not measured on the phone.

5.8 Live partials multiply an offline model's compute

Re-decoding the open utterance every 500 ms takes Parakeet from 0.20–0.23 to 0.52–0.60 of real time, Whisper base from 0.27–0.30 to 0.76–0.88, and Whisper small from 0.52–0.59 to 1.24–1.36. Above 1.0 the decode thread never idles while someone speaks. A final that arrives during a partial has to wait for it: a median of 44–134 ms for Parakeet, and 790–1112 ms for Whisper small. The partial cadence is a dial between how live the text looks and how hot the phone gets.

5.9 The file feed runs ahead of a microphone

The feeder releases each frame at the start of its slot (§3.1). This was known in principle from early on, when the “end of speech” timestamp was corrected, and treated as a small, uniform optimism. The microphone check measured it together with the capture path: 0.2 s in all. A new harness should release each frame at the end of its slot. The published figures are left as measured, so that runs stay comparable.

5.10 The phone under test is also someone's phone

One session run overlapped a video call. It read 0.57 of real time where the per-clip runs read 0.43, with the battery at 38.0 °C, the highest reading of the study, and nothing in the row showed the cause. Clearing the recent apps killed that run and another. The system log showed both, and both runs were discarded. The harness now records screen and call state, and waits out a call before each clip. The second episode also showed the cost of a large model: with about 850 MB resident, opening two apps put the phone into memory thrashing, and the low-memory killer ended 21 cached apps in 17 seconds.

5.11 Integration notes

- Moonshine's SDK runs inference inside `addAudio()`, stalling the caller for the length of a model pass. A live app must not call it from the thread that reads the microphone.
- Moonshine's Spanish streaming models are on the vendor's content network, not on HuggingFace.
- Moonshine's and sherpa-onnx's Android archives both ship `libonnxruntime.so`, in different builds. Picking one at packaging would give one SDK the other's runtime; sherpa-onnx's static-link build avoids the collision.
- sherpa-onnx's Kotlin VAD defaults to a 5 s maximum speech duration, which splits most sentences at a breath.
- The battery temperature an app can read can be minutes old: one reading sat unchanged for 5.5 minutes, then dropped 4 °C in 15 seconds.
- The platform recognizer can return only the last clause of an utterance, and its English language pack had to be installed through the API; the Settings screen did not work.
- On MIUI, an instrumented app cannot open its own activity from the background; opening it through the shell works.

6 Discussion and recommendation

Table 10. The live stacks an app could ship. First text is from speech onset, file-fed; add about 0.2 s for a live microphone. Memory is per model, as an app would load one language at a time.

Live stack	Disk	Spanish WER	English WER, clean / noisy	Final text	First text	Peak memory	Compute, 6 min
Parakeet (one model)	670 MB	4.47%	4.97 / 9.34%	0–350 ms	1.0–1.5 s	1.03–1.11 GB	0.59–0.61
Moonshine small-en + small-es	346 MB	7.24%	7.36 / 8.92%	0–490 ms	0.7–1.0 s	0.64–0.76 GB	0.45–0.70
Moonshine medium-en + small-es	538 MB	7.24%	4.71 / 7.02%	0–770 ms	0.7–1.0 s	0.97 GB	0.45–0.78
Moonshine small-en + platform (es)	224 MB + pack	7.39%	7.36 / 8.92%	0–490 ms	0.7–1.0 s	0.76 GB	0.70 (en)

One model can serve both languages live on this phone, and it is Parakeet. Ship it as the one model, and pad every VAD segment with about half a second of leading silence.

What one model buys is Spanish accuracy. Parakeet makes about 40% fewer Spanish errors than any per-language option, and it serves English within reach of the best: level with Moonshine medium on clean speech, and 2.3 points behind on noisy speech, a gap that padding is expected to close. Over six continuous minutes it used 0.6 of real time with no drift. Its costs are 670 MB of disk, about 1.1 GB resident, which on a 6 GB phone pushes other apps out of memory, and first text 1.0–1.5 s after speech starts. The padding is not optional polish: unpadded, it silently drops whole sentences.

What two models buy is size, first-text speed and heat, not Spanish accuracy. Moonshine small in both languages is half Parakeet’s disk, MIT-licensed, one runtime, the coolest bundled stack, and shows first text 0.2–0.6 s sooner. Its Spanish, however, is no better than the free platform recognizer’s. If disk, memory or heat matter more than Spanish accuracy, it is the choice. Moonshine medium is the most accurate English model, for 192 MB more.

The platform recognizer does not make bundling a model unnecessary. It is free and fast, and its Spanish is as good as Moonshine’s. But it misses one word in four on noisy English, it needs an on-device language pack the user may not have, and it belongs to Google, not to the app.

Beyond the choice of model, a live app built on these results should decode off the capture thread (Moonshine blocks its caller for up to 1.6 s), tune the partial cadence per model rather than leave it at a default, budget for about 1 GB of memory with Parakeet, and expect about 0.2 s more latency from a live microphone than any file-fed figure shows.

7 Limitations

- **One phone.** Timing, memory and heat are properties of this handset and this operating system build.
- **Read speech, one corpus per condition.** Spanish is clean read speech only; there is no noisy Spanish source to match LibriSpeech, and no conversational speech in either language.
- **Sample sizes.** Accuracy rests on 100 clips per source; Whisper and Moonshine tiny were measured on 20. Sessions ran once per language and stack, for six minutes, inside a 10-minute safety cap; a longer session could still find throttling or more memory growth.
- **Padded Parakeet was not measured on the phone.** The recommendation to pad rests on a PC reproduction of the pipeline and on the sentence the session dropped.
- **Stacks, not models.** Runtimes differ, and Moonshine’s SDK exposes no thread setting.
- **Temperatures lag,** and this phone reports no thermal headroom to apps.
- **The microphone check** is one reader, in one room, on one phone. It validates the method; it does not measure accuracy on real voices.

8 Conclusion

On a 2021 mid-range phone, one open model can transcribe English and Spanish live, entirely on-device. Parakeet TDT 0.6b v3 is the most accurate Spanish option measured, competitive in English, and keeps well ahead of real time over continuous speech. It needs 670 MB of disk, about 1 GB of memory, and half a second of silence in front of every segment.

Per-language models buy a smaller, cooler app, not better Spanish. The method itself held up under a live microphone, with one correction: live text arrives about 0.2 s later than a paced file suggests.

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Appendix A Reproducing the results

Everything needed is in the repository at tag v1.0. Weights and audio are not redistributed: scripts fetch them at pinned revisions and rebuild the corpus deterministically. Scoring runs on a PC; the phone emits text and timings only. All scripts are in `scripts/` and run with the project's Python environment.

Step	Command
Corpus	<code>scripts/fetch_corpus.py</code> then <code>scripts/build_corpus.py</code>
Models and runtime	<code>scripts/fetch_models.py --arm B (C, D, E likewise); scripts/fetch_runtime.py</code>
Plumbing, emulator	<code>scripts/run_bench.py --device emulator --plumbing</code>
Short clips, phone	<code>run_bench.py --device physical --arms D:parakeet-tdt-v3-int8 --langs en,es</code>
Sessions	the same, plus <code>--buckets session --sources fleurs_en_norm,fleurs_es --reps 1</code>
Microphone check	<code>run_bench.py --device physical --mic D:parakeet-tdt-v3-int8 --langs en</code>
Score and compare	<code>scripts/summarize.py, scripts/compare.py --standard, scripts/session_report.py</code>
Figures	<code>scripts/plot_tradeoffs.py</code> ; this paper: <code>scripts/build_paper.py</code>

The harness refuses to start unless storage, battery level, temperature and charging state are in range. It checks again before every clip and every 30 s during a session-length clip, waits out calls, and pauses before any clip that would take continuous inference past 10 minutes.

Appendix B Provenance

Artefact	Source	Pinned revision
Moonshine streaming en, base-es	HF moonshine-ai/moonshine-voice-assets, set quantized_26_08_21	0bf2f2e5
Moonshine small-streaming-es	download.moonshine.ai, quantized_26_08_24	SHA-256 per file
Parakeet TDT 0.6b v3 int8	HF csukuangfj/sherpa-onnx-nemo-parakeet-tdt-0.6b-v3-int8	2bda32ec
Whisper small, base	HF csukuangfj/sherpa-onnx-whisper-small, -base	8f3c18b3, bb53ee20
Silero VAD	HF onnx-community/silero-vad	e71cae96
sherpa-onnx	GitHub release 1.13.8, static-link AAR	SHA-256 b22c3fc1...
FLEURS en_us, es_419	HF google/fleurs	70bb2e84
LibriSpeech test-other	HF openslr/librispeech_asr	71cacbf8

Full hashes, file lists and sizes are in `models/MODELS.md` and `corpus/SOURCES.md`. Model licences: MIT (Moonshine streaming, Whisper, Silero VAD), CC-BY-4.0 (Parakeet), Moonshine Community Licence, non-commercial (Moonshine base-es, an experiment only). The corpus sources are CC-BY-4.0.